CS 4720 HW8

1a) Over here, we can compare the arguments on the basis of their positions like if it is A ←→ x, B ←→ y and B ←→ z, so we can say that over here A = x, B = y and B = z and hence the most general unifier in this case can be represented as {x/A, y/B, z/B}, which means that we can substitute x with A and y and z with B and hence both the arguments become P(A,B,B).

b) Let us compare the elements on the basis of their positions over here, so that means y = G(x,x) and y = G(A,B), through which we can say that G(x,x) = G(A,B), which means that A = B = x. This is only possible when A and B are the same constants and not possible when they are different, so that means that there is no possible unifier in this case.

c) Let us compare the elements on the basis of their respective positions again. In this case, we can say that Father(y) = Father(x) and y = John, and if we substitute that in the first equation, then we get Father(John) = Father(x), which means that x = John, and hence the most general unifier in this case can be represented as {Father(y)/Father(x), y/John}, which brings us to a conclusion that both the arguments are same and can represented as Older(Father(y),y).

d) Let us compare the elements on the basis of their position again. In this case, we can say that Father(y) = x and y = x, which, if we substitute in the first equation, then we get y = Father(y), which is not possible because y cannot be its own father, and hence we can say that no unifier is possible in this situation.

2a) i) If we try and unify the first clause with this statement then we can say that Mother(x) = Mother(y), which means that x = y and x = John which means that y = John and we can represent the statement as Ancestor(Mother(John),John) and that gives us the answer.

ii) It will never terminate, since it will never end with John, so it will just generate longer and longer mother-chains everytime and never terminate.

iii) By looking at clause 1, we can say that y's grandmother is an ancestor of their grandmother and the grandmother is an ancestor of their mother, and we can represent it as Ancestor(Mother(Mother(Mother(y),Mother(Mother(y)) and Ancestor(Mother(Mother(y)),Mother(y)), so by looking at clause 2, we can say that Ancestor(Mother(Mother(Mother(y),Mother(Mother(y)) ∧ Ancestor(Mother(Mother(y)),Mother(y)) = Ancestor(Mother(Mother(Mother(y)),Mother(y)), which terminates and gives us the answer.

iv) John's mother cannot be an ancestor of his grandmother, so that's why we can say that it terminates without giving an answer.

b) The resolution algorithm basically takes the opposite of what we are trying to prove, for example, in our case let us assume that Ancestor(John,John) is true, it then adds it to the KB clauses and then later on, it tries to find a contradiction. Now, let us focus on the second step where we will add Ancestor(John,John) to our clauses in KB, so that we can see what output do we get after the algorithm tries to mix all the rules and facts.

Now, we have these clauses:

C1: Ancestor(Mother(x), x)
C2: ¬Ancestor(x, y) ∨ ¬Ancestor(y, z) ∨ Ancestor(x, z) (transitivity that was rewritten in the clause form)
C3: Ancestor(John, John) (our assumption)

Over here, the algorithm basically looks for matching patterns where it finds 1 positive and 1 negative literal that may cancel each other out. Now, let us try resolving C2 with C3. Let us pick up the positive part form C2, which is Ancestor(x,z) and from C3 we have Ancestor(John,John) and then try and unify them after which we get that x = John and z = John too. Now, if we remove the Ancestor(John,John), we get ¬Ancestor(John, y) ∨ ¬Ancestor(y, John), which means that John is not an ancestor of y and y is not an ancestor of John too, then the algorithm checks if any other clause/rule could make us lead to a contradiction, but since the other facts are positive too, we can say that none of the rules lead to a contradiction over here, which means that our assumption didn't break any rules and hence we can say that resolution failed to prove ¬Ancestor(John, John).

3a) For A we can say that every number has some number which is less than or equal to it and for B, we can say that there exists some number such that it is less than or equal to every other number.

b) It is true since if we assume y = 0, then there will always be a number that is less than or equal to any x, since 0 is the smallest number.

c) It is true since if we assume y = 0, then it will be the number which will be less than or equal to every other x since 0 is the smallest number.

d) Since A says "Every number has some number less than or equal to it" and B says that "There exists one number that is less than or equal to every other number", so which means that both of them are true because of 0, as it is less than or equal to every other number, so we can say that (A) entails (B) in natural numbers.

e) Since we are only assuming natural numbers, we can say that there exists a number that is less than or equal to everything because of 0. Now, since A is true as well, for any number, say x, we can always pick 0 and say that x is always greater than or equal to it; hence, we can say that B entails A in the natural numbers.

f) Let us try and prove this problem with the help of resolution. Let us first write everything in the form of logical statements, which means that for B, we can say that there exists a y such that everything is greater than or equal to say some constant “a”. Then we can represent the clause as ge(x, a) (where ge just means that it is greater than or equal to), and where ¬A that there is some x which has no smaller or equal y, so in this case we can say there exists some constant “b”, such that there exists no such number which is less than or equal to it and hence we can represent its clause as ¬ge(b,y).

Now, let us try to unify ge(x, a) and ¬ge(b,y), which means that x = b and y = a, and after substituting the 2 clauses may cancel each other completely, and we get an empty clause, which means that we have a contradiction and hence we can say that the assumption “(B)” does not imply “(A)”, where (A) follows from “(B)”.

Hence, we have proved that assumption (B) and not (A) is impossible, so that means that (B) does logically imply (A) and (A) follows from (B).

g) Let us first write both (A) and (B) in the form of logical statements, so now (A) means that for every x, there exists a y such that x is always greater than or equal to y, which means that its clause can be represented as ge(x,f(x)) (meaning x is greater than or equal to f(x)) and now we take the opposite of B, which is ¬B, which says that for every y there exists an x such that it is not greater than or equal to y, which means that its clause can be represented as ¬ge(g(y),y) (meaning that for every y there is always exists some g(y) which is not greater than or equal to y).

Now, let us unify ge(x,f(x)) and ¬ge(g(y),y), and after doing that we get x = g(y) and f(x) = y, so we can substitute x as g(y) in f(x) = y, which makes it f(g(y)) = y, which is not possible since y would become equal to a term that contains itself.

Now, since we can’t fit these 2 statements together, we can’t derive a contradiction, and hence we can say that this proof fails.